Amirreza Ahadzadeh

Software Engineer · B.Sc. Computer Science (Final Year)

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github.com/amilion · portfolio-website-five-kappa-31.vercel.app

SUMMARY

Final-year CS student at the University of Tehran (GPA 19.0/20) and software engineer who builds systems from the ground up. Strong in **Python, C++, and C** with solid algorithmic foundations; experienced implementing ML models and algorithms from scratch, building full-stack REST apps, and shipping with testing and CI/CD on Linux.

TECHNICAL SKILLS

Languages: Python (advanced), C++, C, SQL, TypeScript / JavaScript

Backend & APIs: Django, Django REST Framework, REST API design, JWT authentication, SQLAlchemy

Frontend: React, TypeScript, React Hook Form, Axios

ML & Scientific: PyTorch, NumPy, spiking neural networks (CoNeX, PyMonNTorch)

Databases & Tools: PostgreSQL, Git, GitHub Actions (CI/CD), pytest, Swagger / OpenAPI, Linux

Foundations: Data structures & algorithms, dynamical systems, data mining, probability, linear algebra

PROJECTS

Communicating World Models — from Scratch — Python, PyTorch GitHub

- Built multi-agent world-model RL agents from scratch in PyTorch: a **VAE** encodes observations into a latent space and a recurrent state-space model (RSSM) learns environment dynamics for latent "imagination" (Dreamer-style).
- Implemented **actor–critic** behavior learning with λ-returns and symlog/symexp value transforms, plus a **broadcast communication module** and a shared critic that let agents exchange messages and predict the best next state.

XGBoost — from Scratch — Python, NumPy GitHub

- Implemented gradient-boosted regression trees from scratch in NumPy, following the XGBoost paper (Chen & Guestrin), with a scikit-learn-compatible GradientBoostingRegressor API.

Full-Stack E-Commerce REST Platform — Django REST Framework, React, PostgreSQL GitHub

- Designed a modular RESTful e-commerce backend in Django REST Framework spanning 15+ apps (products, carts, orders, discounts, shops, comments, and more).
- Implemented stateless **JWT auth with token blacklisting** and custom permissions; modeled PostgreSQL data (hierarchical categories, filtering); documented the API with Swagger/OpenAPI; added a **GitHub Actions CI pipeline** (migrations + tests) and a typed React + TypeScript frontend.

Print-Job Telegram Bot — Python, SQLAlchemy, PostgreSQL GitHub

- Delivered a production Telegram bot for a real client that connects customers with a seller: customers submit orders, the seller prices each one, and the bot manages payment-receipt confirmation and completion notifications.
- Built with **python-telegram-bot** and a modular architecture (conversation/command handlers, permission decorators) over SQLAlchemy ORM models on PostgreSQL.

Hippocampal CA3 Short-Term Memory — Python, PyTorch, CoNeX, PyMonNTorch GitHub

- Built a spiking-neural-network model of hippocampal CA3 that performs pattern completion (recalling full patterns from partial cues), validating biological plausibility through oscillatory network dynamics.

RESEARCH EXPERIENCE

Research Assistant — Goethe Research Experience Program (GREP) Apr 2026 – Present

Prof. Dr. Jochen Triesch's Cognition Lab, Goethe University · Frankfurt, Germany

- Investigating mechanisms for introducing controlled self-disorganization into recurrent spiking neural network (rSNN) architectures.

Research Assistant — Computational Neuroscience Research Lab (CNRL) Jul 2024 – Present

Prof. Dr. Mohammad Ganjtabesh · University of Tehran, Iran

- Developed the lab's first spiking **Cortical Column for object recognition** (Python/PyTorch/CoNeX): a feature-binding mechanism via R-STDP and dynamic reference frames that build complete object models, with bio-plausible plasticity rules. (repo)

TEACHING

Teaching Assistant — University of Tehran Feb 2024 – Present

- Algorithm Design & Analysis, Data Structures & Algorithms, and Calculus I & II.

EDUCATION & HONORS

B.Sc. Computer Science — University of Tehran 2022 – 2026 (expected)

GPA 19.0/20 · Top 0.3% national entrance exam (full tuition waiver) · Top student, 7 consecutive semesters.